NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM I

MH2100 – Calculus III

September 2023 TIME ALLOWED: 90 minutes

Name:

Matric. no.: Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. **ONE** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1. **(17 marks)**

(i) [3 marks] Find parametric equations for the tangent line at the point when $t = 1$ to the curve with parametric equations

$$x = 3t^2, \quad y = 5t, \quad z = 7t^3.$$

(ii) [3 marks] The cartesian equation $y = ax + b$ describes the line tangent to the curve

$$\begin{cases} x(t) = e^t \\ y(t) = t^2 + 2t - 7 \end{cases}$$

at the point defined by $t = 0$. Find the values of a and b.

(iii) [3 marks] Find all points (P, Q) on the following curve where the tangent line is vertical.

$$x = t^3 - 9t + 18, \quad y = \sin(t)$$

(iv) [3 marks] At the point $(0, \pi, 1)$, find the directional derivative of the function $f(x, y, z) = \sin(xyz)$ in the direction of the vector $\langle 1, 2, 2\rangle$.

(v) [5 marks] Find all unit direction vectors $\mathbf{u} = \langle a, b\rangle$ in which the directional derivative of the function $f(x, y) = ye^{-xy}$ at the point $(0, 2)$ has the value 1.

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QUESTION 2. **(16 marks)**

(i) [4 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y)\to(1,0)} \frac{\sin(xy)}{x^3y+(x-y)^4}$$

(b)

$$\lim_{(x,y)\to(0,0)} \frac{x^3y}{x^6+y^2}$$

(ii) [5 marks] Find all points (P,Q,R) of the level surface given by

$$x^3+3x^2y+6yz-3z^2=7$$

where the tangent plane has an equation of the form $Ax+Ay=B$ for some constants A and B.

(iii) [7 marks] Define

$$f(x,y)=\begin{cases} \frac{y^3}{x^2+y^2}; & \text{if } x^2+y^2\neq 0\\ 0; & \text{if } x^2+y^2=0. \end{cases}$$

(a) Is f continuous at $(0,0)$? Justify your answer.

(b) Find $f_x(0,0)$ and $f_y(0,0)$.

(c) Is f differentiable at $(0,0)$? Justify your answer.

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QUESTION 3. **(17 marks)**

(i) [3 marks] Find the second order partial derivative w_{xy} where

$$w = y\sin(3x^2 - 5y).$$

(ii) [3 marks] Use the Chain Rule to find $\dfrac{\partial u}{\partial y}$ at $(x, y, z) = (-1, 0, 1)$ for the function

$$u(p, q, r) = \frac{p^2}{q + r},$$

where

$$p = x + 2y, \quad q = y + 2z, \quad r = x + y + z.$$

(iii) [3 marks] Assume that z is uniquely determined in terms of x and y by the equation

$$z^3 - xy + yz + y^3 = 2.$$

Find the value of $\dfrac{\partial z}{\partial y}$ at the point $(1, 1, 1)$.

(iv) [3 marks] In which directions is the directional derivative of

$$f(x, y) = \sin(xy) - x^2 + xy$$

at $(\pi, 0)$ equal to zero? Write the unit vector for each direction.

(v) [5 marks] Let

$$f(x, y) = x^3 - y^2 + xy - \frac{x}{2}.$$

(a) Find all critical points of $f(x, y)$.

(b) For each critical point found in Part (a), determine whether or not it is a local max/min or a saddle point.

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